

AYMANE AIT DADS

AI Engineer Intern | RAG · Agentic Frameworks · Machine Learning

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by engineering an end-to-end LoRA SFT pipeline on a 30B-parameter Mamba-Transformer with BF16 precision and 8,192-token context. Reduced manual network analysis time by 60% at Orange Maroc by deploying Random Forest and K-Means models on 100K+ production records, translating model outputs directly into stakeholder maintenance recommendations. Brings hands-on experience in agentic pipelines, RAG systems, LLM orchestration, and machine learning across both research and production settings - directly matching the Engineering and Research team mandate described in this role. Aims to contribute production-ready AI solutions at Magellan Technology Research Institute by applying ablation-driven ML engineering and agentic framework design to real-world challenges.

CORE COMPETENCIES

RAG | Agentic Frameworks | Machine Learning | LLM Orchestration | AI Solutions | Production-Ready Projects | Research & Engineering | Hands-on AI Building

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built end-to-end **machine learning pipeline** combining Random Forest and K-Means models on 100K+ fiber-optic network records to classify quality across thousands of network nodes.
- Developed clustering pipeline that mapped customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for production use.
- Reduced manual analysis time by **~60%** through automated feature engineering, model evaluation workflows, and reusable Pandas/Scikit-learn preprocessing scripts.
- Delivered stakeholder presentations translating **model outputs into actionable maintenance recommendations**, bridging data science findings and business decision-making for non-technical audiences.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context).
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers for verified chain-of-thought data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection - NTIRE 2026 @ CVPR

EURECOM ImSecu Course + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** (private Kaggle leaderboard, 0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M params) with a custom MLP head, dual learning rates, and FP16 mixed precision on 250K images across 25+ generator types.
- Submitted to **NTIRE 2026 @ CVPR CodaBench** international benchmark; achieved best AUC of 0.793 with 1-epoch training via 3-model ensemble and 10-view TTA, demonstrating superior out-of-distribution generalization over longer training runs.

PyTorch · OpenCLIP · ViT · AdamW · Kaggle GPU

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Hugging Face Transformers | Unsloth | TRL | PEFT | Scikit-learn | vLLM

LLM & Agentic AI: LoRA SFT | RAG | Agentic Pipelines | LLM Orchestration | LangChain | Prompt Engineering | Chain-of-Thought

Data Science & MLOps: Python | Pandas | NumPy | Feature Engineering | BF16/FP16 Mixed Precision | Git | Power BI